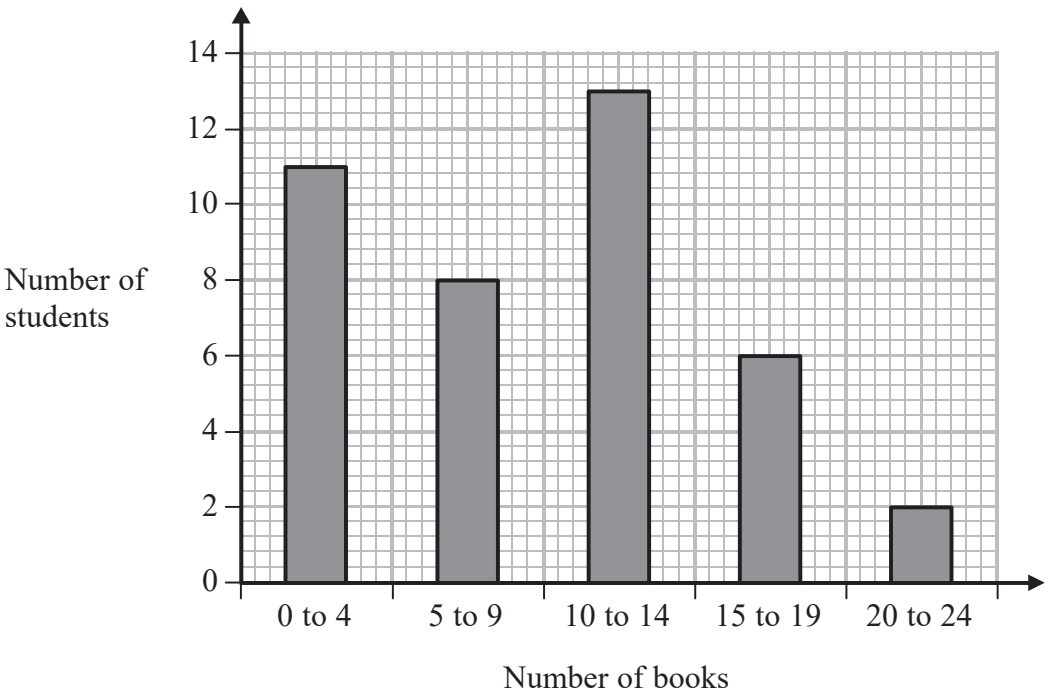
4 Fran asks each of 40 students how many books they bought last year.

The chart below shows information about the number of books bought by each of the 40 students.



(a) Work out the percentage of these students who bought 20 or more books.

20 or more: 2 students out of 40

	Students	Percent	
	40	100	
÷20	2	5	÷20

1 mark
Use the final bar: 2 out of 40.

1 mark
Correct percentage.

5 %

(2)

(b) Show that an estimate for the mean number of books bought is 9.5
You must show all your working.

Interval	Frequency	Midpoint	f × x
0–4	11	2	22
5–9	8	7	56
10–14	13	12	156
15–19	6	17	102
20–24	2	22	44

1 mark
Frequencies and midpoints.

1 mark
Multiply each frequency by its midpoint.

$\Sigma f = 40$ $\Sigma fx = 380$

$$\text{Estimated mean} = \frac{380}{40} = 9.5 \text{ books}$$

1 mark
Divide the sum of fx by the total frequency.

1 mark
Correct figures show the stated mean of 9.5 books.

(4)

(Total for Question 4 is 6 marks)